```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

from sklearn import metrics

train = pd.read_csv("train.csv")
test = pd.read_csv("test.csv")

print(train.head())
print(test.head())

X = train.drop("label", axis=1)
y = train["label"]

print (X. shape)
print(y.shape)

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.3, random_state=42)

print (X_train.shape)
print(X_test. shape)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

rbf_model = SVC(kernel="rbf")
rbf_model.fit(X_train, y_train)

print("SVM model trained successfully")

y_rbf_pred = rbf_model.predict(X_test)

print("Predicted values:", y_rbf_pred[10:15])
print("Actual values:", y_test.iloc[10:15].values)
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acc_rbf = metrics.accuracy_score(y_test, y_rbf_pred)

print("Accuracy:", "{:.2f}".format(acc_rbf * 100), "%")

test_scaled = scaler.transform(test)
test_predictions = rbf_model.predict(test_scaled)

print("Predicted labels:", test_predictions[:10])

for i in np.random.randint(@, len(X_test), 4):
two_d = np.reshape(X_test[i], (28, 28))
plt.title('Predicted label: {}'.format(y_rbf_pred[i]))
plt.imshow(two_d, cmap='gray')
plt.show()
```